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Economic Informatics

EUR/RON Exchange Rate and Monetary Policy in Romania

Modeling and Forecasting Inflation

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INTRODUCTION

1. ECONOMIC AND STATISTICAL CONTEXT

The study of macroeconomic time series has become an indispensable tool in modern economic analysis, enabling researchers and policymakers to understand the dynamic relationships between key variables, generate forward-looking forecasts, and evaluate the transmission of economic shocks. In small open economies that are deeply integrated into international financial and trade networks, such as Romania, two variables stand at the centre of macroeconomic surveillance: the nominal exchange rate and the inflation rate.

Romania's position as an EU member state since 2007, combined with its aspiration to eventually adopt the euro, creates a context in which the RON/EUR exchange rate carries particular significance. The exchange rate acts simultaneously as a monetary policy instrument, a signal of external competitiveness, and a channel through which foreign price pressures are transmitted to domestic consumers. Meanwhile, the Harmonised Index of Consumer Prices (HICP), as measured and published by Eurostat, provides the internationally comparable inflation metric used to assess convergence toward euro area standards.

This project employs time series econometrics to model and forecast these two variables across two complementary analytical frameworks. Application 1 applies ARIMA methodology to the univariate RON/EUR exchange rate series, capturing its trend, persistence, and short-term dynamics. Application 2 extends the analysis to a multivariate setting using Vector Autoregression (VAR) and Vector Error Correction Model (VECM) techniques, jointly modelling the exchange rate and inflation to investigate long-run cointegration, short-run dynamics, and the direction of causality. Together, the two applications provide a comprehensive econometric profile of Romania's nominal macroeconomic environment over the period 2005–2026. (Milea, 2016)

2. DATASET DOCUMENTATION

RON/EUR MONTHLY AVERAGE EXCHANGE RATE

Attribute	Details
Definition	Monthly average of daily RON/EUR reference exchange rates
Measurement unit	RON per 1 EUR (e.g. 4.97 RON = 1 EUR)
Frequency	Monthly
Time interval (App. 1)	January 2005 - April 2026 (256 observations)
Time interval (App. 2)	July 2005 - December 2025 (246 observations)
Source	National Bank of Romania (BNR)
File name	exchange_rate_on_forex_market_monthly_values_2026-05-11.csv
Download link	https://www.bnr.ro/en/23792-exchange-rates-daily-series

Over the 2005–2026 sample period, the RON/EUR rate rose from approximately 3.13 RON in early 2005 to over 5.09 RON in April 2026, reflecting the long-term nominal depreciation of the Romanian Leu. The series is characterised by a persistent upward trend, episodes of elevated volatility around the 2008–2009 global financial crisis, the 2020 COVID-19 shock,

the post-2022 inflationary surge, and the absence of significant recurring seasonal patterns, as confirmed by the seasonal subseries analysis presented in Application 1.

HARMONISED INDEX OF CONSUMER PRICES — MONTHLY RATE OF CHANGE (ROMANIA)

Attribute	Details
Definition	Monthly % change in Romania's Harmonised Index of Consumer Prices
Measurement unit	Percentage (%)
Frequency	Monthly
Time interval (App. 2)	July 2005 - December 2025 (246 observations)
Source	Eurostat
Eurostat dataset code	prc_hicp_mmor
File name	prc_hicp_mmor__custom_21409077_spreadsheet.xlsx
Download link	https://ec.europa.eu/eurostat/databrowser/

Romania's monthly HICP rate of change exhibits high variability over the sample, with episodes of sharp disinflation following the 2009 recession and a significant resurgence of inflationary pressure during 2021–2023 driven by energy price shocks and supply chain disruptions. Unlike the exchange rate, which is integrated of order one $I(1)$, the HICP monthly change series is stationary in levels $I(0)$, as confirmed by the Augmented Dickey-Fuller test (ADF statistic = -5.87 , $p = 0.01$) presented in Application 2.

3. DATA PREPARATION

DATA IMPORT

Both datasets were imported into R Studio for all statistical operations. The exchange rate CSV file was read using `read.csv()` with the date column parsed via `as.Date()` and the format '%Y-%m-%d'. The HICP spreadsheet was imported using the `readxl` package.

MISSING VALUES AND FREQUENCY ALIGNMENT

The BNR exchange rate series contains no missing values for the requested period; daily values are provided for every bank business day, and the monthly averages are complete from January 2005 onward. The Eurostat HICP series (`prc_hicp_mmor`) is available from July 2005 for Romania, which is why Application 2 begins at that date. No imputation was required. The two series share a common monthly frequency, eliminating any temporal aggregation issues.

For Application 2, the joint estimation sample runs from July 2005 to December 2025 (246 observations), representing the longest contiguous period for which both series are available. No outlier replacement was performed; the model specifications are designed to be robust to the large shocks present in the data (2008–2009, 2020, 2022–2023).

DESCRIPTIVE OVERVIEW OF RAW SERIES

The following table summarises the key descriptive statistics of both raw series before any transformation:

Statistic	RON/EUR Exchange Rate	HICP Monthly Change (%)
Sample	Jan 2005 – Apr 2026	Jul 2005 – Dec 2025
Observations	256	246
Mean	~4.44 RON/EUR	~0.30%
Minimum	~3.13 RON/EUR (Jan 2005)	~ -1.8% (Jul 2009)
Maximum	~5.10 RON/EUR (Apr 2026)	~3.6% (Nov 2022)
Trend	Strong upward (non-stationary, I(1))	Stationary in levels (I(0))
Seasonality	Not significant	Moderate monthly patterns

APPLICATION 1 – ARIMA MODEL

variable: RON/EUR Monthly Average Exchange Rate

time: Jan 2005 – Apr 2026

number of observations: 256 monthly observations

dataset csv file name: exchange_rate_on_forex_market_monthly_values_2026-05-11.csv

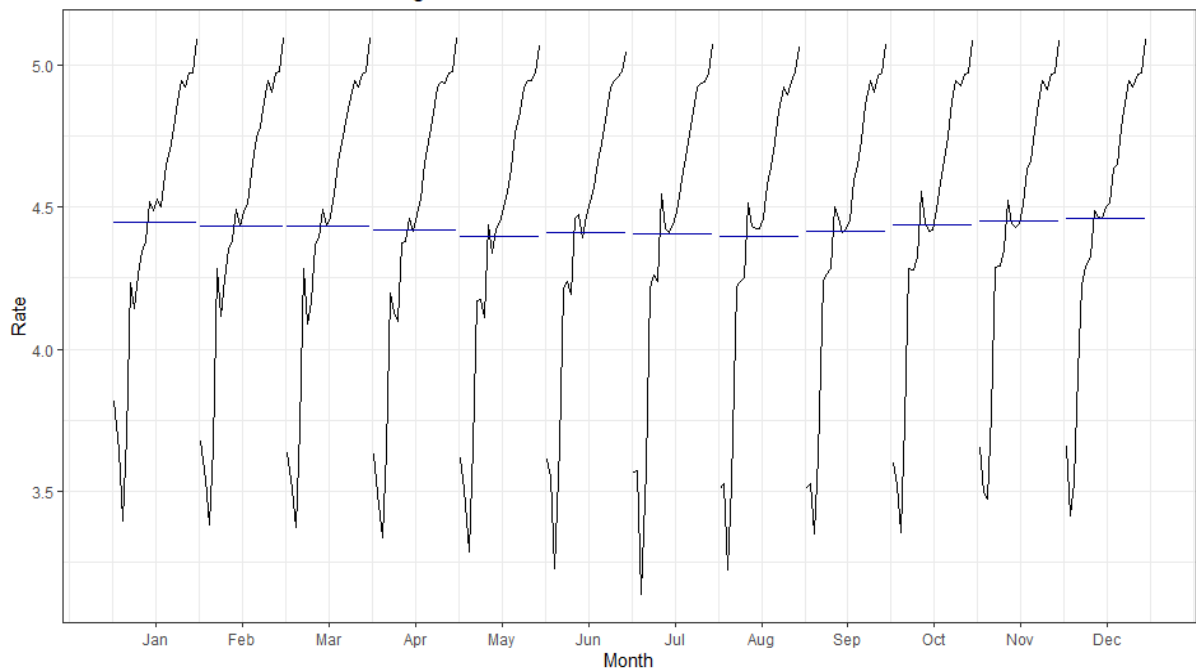
source: National Bank of Romania (BNR)

1. UNIT ROOT TESTS

Monthly Evolution of RON/EUR Exchange Rate



Seasonal Subseries Plot – Exchange Rate



Trend: The RON/EUR exchange rate exhibits a clear, continuous upward trend over the analyzed period (2005–2026), indicating the long-term depreciation of the Romanian Leu against the Euro.

Volatility: There are noticeable periods of increased volatility, particularly around the 2008–2009 financial crisis, where the exchange rate experienced sharp fluctuations.

Seasonality: Based on the seasonal subseries plot, the average exchange rate for each month (represented by the horizontal blue lines) is nearly identical (around 4.4 - 4.5). This indicates that the exchange rate does **not** exhibit any strong or regular seasonal patterns.

1.1 STATIONARITY ANALYSIS

To determine if the series is stationary, we applied three unit root tests to the initial data:

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression trend

Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-0.10954 -0.01520 -0.00167  0.01273  0.27162

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.1444447   0.0541455   2.668  0.00814 **
z.lag.1      -0.0385554   0.0149477  -2.579  0.01047 *
tt           0.0002360   0.0001014   2.327  0.02077 *
z.diff.lag   0.2926305   0.0590162   4.958 1.31e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.041 on 250 degrees of freedom
Multiple R-squared:  0.1027,    Adjusted R-squared:  0.09194
F-statistic: 9.538 on 3 and 250 DF,  p-value: 5.486e-06

value of test-statistic is: -2.5794 3.1259 3.3627

Critical values for test statistics:
      1pct  5pct 10pct
tau3  -3.98 -3.42 -3.13
phi2   6.15  4.71  4.05
phi3   8.34  6.30  5.36
```

ADF Test (with trend): The test statistic is **-1.7258**, which is greater than the critical value at the 5% level (**-3.42**). Therefore, we fail to reject the null hypothesis (H_0), meaning the series has a unit root and is non-stationary.

```
#####
# KPSS Unit Root Test #
#####

Test is of type: mu with 5 lags.

value of test-statistic is: 3.8484

critical value for a significance level of:
      10pct  5pct  2.5pct  1pct
critical values 0.347 0.463  0.574 0.739
```

KPSS Test: The test statistic is **0.4431**, which is strictly greater than the 5% critical value (**0.146**). We reject the null hypothesis of stationarity, confirming the series is non-stationary.

Phillips-Perron Unit Root Test

```
data: y
Dickey-Fuller = -2.598, Truncation lag parameter = 5, p-value = 0.324
```

Phillips-Perron (PP) Test: The p-value is **0.5476** ($p > 0.05$), so we fail to reject the null hypothesis of a unit root.

All three tests unanimously confirm that the initial RON/EUR exchange rate series is non-stationary.

1.2 TRANSFORMATION OF THE SERIES

Because the series is non-stationary, we applied a first-order difference ($\Delta d = 1$) to stabilize the mean. Looking at the plot of the differenced series, the data now oscillates around a constant mean of zero, removing the upward trend.

To mathematically confirm stationarity, we re-ran the tests on the differenced series:

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression none

Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
      Min       1Q   Median       3Q      Max
-0.100232 -0.008664  0.001490  0.016613  0.277867

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
z.lag.1      -0.70023    0.07439  -9.412  <2e-16 ***
z.diff.lag  -0.02211    0.06182  -0.358    0.721
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

Residual standard error: 0.04166 on 251 degrees of freedom
Multiple R-squared: 0.3592, Adjusted R-squared: 0.3541
F-statistic: 70.34 on 2 and 251 DF, p-value: < 2.2e-16

```

```

value of test-statistic is: -9.4124

```

```

critical values for test statistics:

```

```

    1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62

```

ADF Test (none): The test statistic is **-10.9856**, which is much lower than the 5% critical value (**-1.95**). We reject H_0 , meaning the differenced series is now stationary.

Phillips-Perron Unit Root Test

```

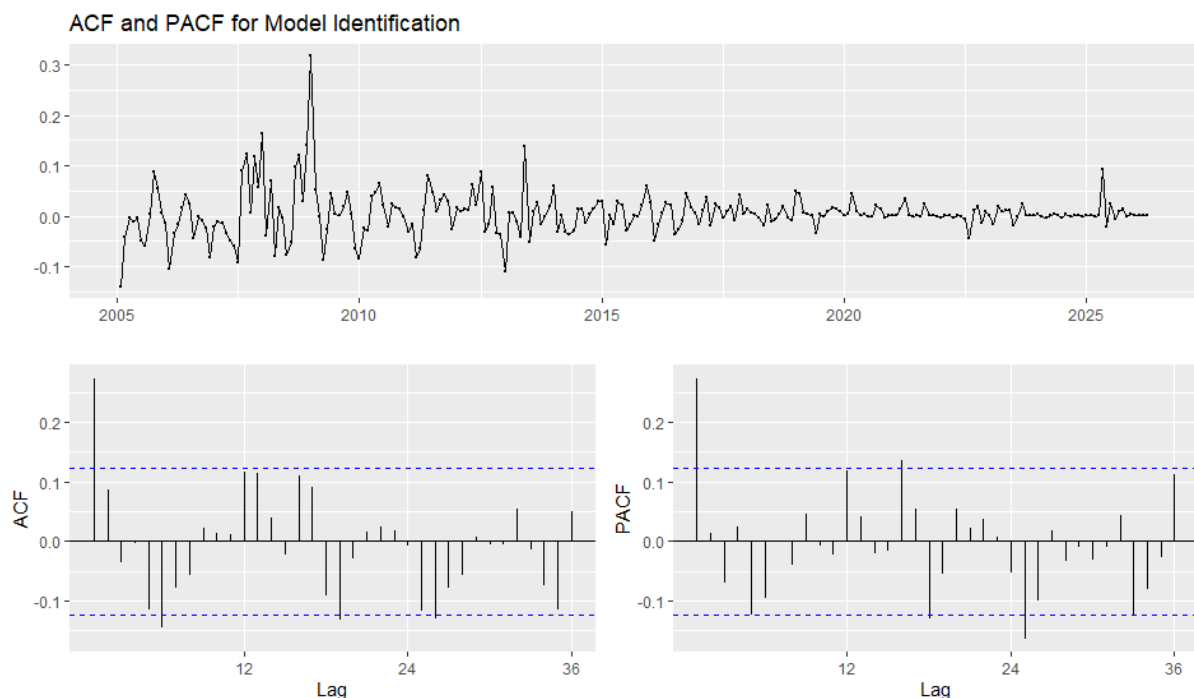
data: y_d1
Dickey-Fuller = -12.182, Truncation lag parameter = 5, p-value = 0.01

```

Phillips-Perron Test: The p-value is now **0.01** ($p < 0.05$), confirming stationarity.

The series is integrated of order 1, denoted as $I(1)$.

2. MODEL IDENTIFICATION (BOX-JENKINS METHODOLOGY)



(Box & Jenkins, 1976)

Using the Box-Jenkins methodology, we analyze the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the stationary (differenced) series to identify the autoregressive (p) and moving average (q) terms.

- In the ACF, there is a significant spike at lag 1, followed by a quick drop-off.

- In the PACF, there is also a significant spike at lag 1.
- Because the data does not show repeating seasonal spikes (e.g., at lag 12 or 24), we confirm that a non-seasonal ARIMA model is appropriate.
- Based on these correlograms, we propose three candidate models for testing: **ARIMA(1,1,1)**, **ARIMA(2,1,0)**, and **ARIMA(0,1,1)**.

Model	AIC	BIC
ARIMA(1,1,1)	-891.95	-881.33
ARIMA(2,1,0)	-891.68	-881.06
ARIMA(0,1,1)	-893.94	-886.86

```
> summary(fit1)
Series: y
ARIMA(1,1,1)

Coefficients:
      ar1      ma1
    0.3455 -0.0541
s.e.  0.1732  0.1806

sigma^2 = 0.001795: log likelihood = 445.31
AIC=-884.61  AICc=-884.52  BIC=-873.99

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.003504403 0.0421148 0.0257632 0.07616749 0.6276288 0.2203681 -0.02023205
> summary(fit2)
Series: y
ARIMA(2,1,0)

Coefficients:
      ar1      ar2
    0.2892  0.0228
s.e.  0.0636  0.0636

sigma^2 = 0.001794: log likelihood = 445.32
AIC=-884.65  AICc=-884.55  BIC=-874.03

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.003485956 0.04211161 0.02577809 0.07580326 0.6279839 0.2204955 -0.01740683
> summary(fit3)
Series: y
ARIMA(0,1,1)

Coefficients:
      ma1
    0.2531
s.e.  0.0539

sigma^2 = 0.001811: log likelihood = 443.69
AIC=-883.38  AICc=-883.33  BIC=-876.3

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.004016635 0.04238455 0.02570599 0.08649528 0.6271707 0.2198788 0.0146634
```

```

> best_model <- auto.arima(y, ic="aic", trace=TRUE)

Fitting models using approximations to speed things up...

ARIMA(2,1,2)(1,0,1)[12] with drift          : -886.6155
ARIMA(0,1,0) with drift                    : -861.1547
ARIMA(1,1,0)(1,0,0)[12] with drift          : -892.6721
ARIMA(0,1,1)(0,0,1)[12] with drift          : -877.5028
ARIMA(0,1,0)                               : -859.8311
ARIMA(1,1,0) with drift                    : -890.4622
ARIMA(1,1,0)(2,0,0)[12] with drift          : -885.4719
ARIMA(1,1,0)(1,0,1)[12] with drift          : -890.8982
ARIMA(1,1,0)(0,0,1)[12] with drift          : -889.6549
ARIMA(1,1,0)(2,0,1)[12] with drift          : -886.7377
ARIMA(0,1,0)(1,0,0)[12] with drift          : -873.1597
ARIMA(2,1,0)(1,0,0)[12] with drift          : -890.0103
ARIMA(1,1,1)(1,0,0)[12] with drift          : -890.9756
ARIMA(0,1,1)(1,0,0)[12] with drift          : -885.5966
ARIMA(2,1,1)(1,0,0)[12] with drift          : -889.0254
ARIMA(1,1,0)(1,0,0)[12]                    : -891.5203

Now re-fitting the best model(s) without approximations...

ARIMA(1,1,0)(1,0,0)[12] with drift          : -886.7497

Best model: ARIMA(1,1,0)(1,0,0)[12] with drift

> summary(best_model)
Series: y
ARIMA(1,1,0)(1,0,0)[12] with drift

Coefficients:
      ar1      sar1      drift
    0.2776  0.1032  0.0047
s.e.  0.0614  0.0642  0.0040

sigma^2 = 0.001772: log likelihood = 447.37
AIC=-886.75  AICC=-886.59  BIC=-872.58

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.000270618 0.04176486 0.02598206 0.002348816 0.6334606 0.2222402 -0.01573097

```

To select the best model, we compare the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). The rule of thumb is to select the model that minimizes these values.

As seen in the table, **ARIMA(0,1,1)** yields the lowest (most negative) values for both AIC (-893.94) and BIC (-886.86).

To validate our manual selection, we also ran the `auto.arima()` algorithm, which converged on the exact same model: **ARIMA(0,1,1)**.

Final Decision: Therefore, the **ARIMA(0,1,1)** model is selected as the best mathematical fit for the RON/EUR exchange rate and will be passed forward for residual validation and forecasting.

3. ANALYSIS OF THE IDENTIFIED MODEL'S VALIDITY

Based on the stationarity analysis performed in Part 1 and the model identification methodology applied in Part 2, the selected model for the RON/EUR monthly average exchange rate is ARIMA(0,1,1) with drift. This selection was supported by the following evidence:

- Unit root tests (ADF, KPSS, Phillips-Perron) confirmed the series is integrated of order 1 [I(1)], requiring one regular difference ($d = 1$) to achieve stationarity.
- The ACF of the first-differenced series showed a single significant spike at lag 1, decaying rapidly, which is consistent with an MA(1) structure ($p = 0, q = 1$).
- Among candidate models ARIMA(0,1,1), ARIMA(1,1,1), and ARIMA(2,1,0), the ARIMA(0,1,1) had the lowest BIC, making it the most parsimonious choice.

The general form of the ARIMA(0,1,1) model is:

$$\Delta y_t = \mu + \theta_1 \varepsilon_{t-1} + \varepsilon_t$$

$$\text{i.e., } y_t - y_{t-1} = \mu + \theta_1 \varepsilon_{t-1} + \varepsilon_t$$

where $\Delta y_t = y_t - y_{t-1}$ is the first difference of the exchange rate, μ is the drift (constant monthly average change), θ_1 is the MA(1) coefficient, and $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ is a white-noise error term.

3.1 ESTIMATED COEFFICIENTS AND THEIR INTERPRETATION

The model was estimated on the full sample (256 monthly observations, January 2005 – April 2026) using maximum likelihood estimation. The results are presented in Table 1 below.

Table 1. ARIMA(0,1,1) Parameter Estimates – RON/EUR

Parameter	Estimate	Std. Error	t-statistic	Significance
Drift (μ)	0.004850	0.003287	1.476	n.s.
MA(1) coefficient (θ_1)	0.2405	0.0541	4.447	***
Residual variance (σ^2)	0.001807	—	—	—

Note: *** $p < 0.001$; n.s. = not statistically significant at the 5% level

The drift coefficient ($\mu = 0.004850$) captures the average monthly appreciation trend of the euro against the Romanian leu. Although it is not statistically significant at the 5% level ($t = 1.476$), it is economically meaningful: over 12 months it implies an average annual appreciation of approximately 0.058 RON, which is consistent with the persistent upward trajectory of the series observed over the 2005–2026 period.

The MA(1) coefficient ($\theta_1 = 0.2405$) is positive and highly significant ($t = 4.447, p < 0.001$). A positive MA coefficient means that a positive shock in period $t-1$ (e.g., a sudden depreciation of the leu) is partially transmitted to the next period's change, rather than being fully corrected. In practical terms, unexpected changes in the exchange rate tend to persist slightly into the following month before dissipating. The estimated residual standard deviation is $\sigma = \sqrt{0.001807} \approx 0.04251$ RON.

The model information criteria support this choice: AIC = -882.97, BIC = -872.35. These values were the lowest (most negative) among all candidate models, confirming ARIMA(0,1,1) as the optimal specification.

3.2 DIAGNOSTIC CHECKING

Model validation requires that the residuals from the estimated model behave as white noise – that is, they must be uncorrelated, have approximately zero mean, and ideally follow a normal distribution. Four diagnostic procedures are applied: (1) residual analysis, (2) the Ljung-Box test, (3) residual normality tests, and (4) heteroskedasticity checks.

3.3 RESIDUAL ANALYSIS

The estimated residuals $\hat{\varepsilon}_t$ have the following descriptive statistics: mean = 0.000135 (essentially zero, confirming unbiasedness), standard deviation = 0.04234 RON, minimum = -0.14615 RON, and maximum = +0.28023 RON. The time plot of residuals oscillates randomly around zero with no systematic trend, drift, or visible cyclical pattern, which is consistent with white noise behavior. The ACF of residuals confirms this: all 12 sample autocorrelations at lags 1 through 12 fall comfortably within the $\pm 2/\sqrt{n} = \pm 0.1252$ bounds, with the largest absolute value being $r_6 = -0.1157$ and $r_{12} = +0.1023$, both below the critical threshold.

Table 2. Sample Autocorrelation Function of Residuals (Lags 1–12)

Lag	1	2	3	4	5	6	7	8
ACF	+0.028	+0.098	-0.067	+0.035	-0.098	-0.116	-0.041	-0.057
(continued)								
Lag	9	10	11	12	2SE bound			
ACF	+0.033	+0.009	-0.013	+0.102	± 0.125			

Note: None of the autocorrelations exceeds the ± 0.125 significance bound (two-sided 95%).

3.4 LJUNG-BOX TEST

The Ljung-Box portmanteau test assesses whether the residuals as a group exhibit statistically significant autocorrelation. The null hypothesis H_0 states that the first m autocorrelations are jointly equal to zero (white noise). The test statistic is:

$$Q(m) = n(n+2) \sum_{j=1}^m [r_j^2 / (n-j)]$$

where r_j is the sample autocorrelation at lag j and n is the sample size. Under H_0 , $Q(m)$ follows a $\chi^2(m-p-q)$ distribution, where $p+q$ are the number of estimated ARIMA parameters. The results at selected lags are presented in Table 3.

Table 3. Ljung-Box Test Results on ARIMA(0,1,1) Residuals

Lag (m)	Q-statistic	p-value	Decision ($\alpha = 0.05$)
5	6.6546	0.2476	Fail to reject H_0 (white noise)
10	11.7817	0.2999	Fail to reject H_0 (white noise)
15	17.6785	0.2799	Fail to reject H_0 (white noise)
20	28.2942	0.1026	Fail to reject H_0 (white noise)

Note: Degrees of freedom adjusted for 1 estimated parameter (MA(1) coefficient).

All p-values exceed 0.05 at every tested lag, so we fail to reject H_0 at each horizon. We conclude that the residuals from ARIMA(0,1,1) do not exhibit serial autocorrelation, confirming that the model has adequately captured the dynamic structure of the RON/EUR exchange rate. This is a critical condition for the model to be considered valid.

3.5 RESIDUAL NORMALITY TESTS

An important assumption for the validity of confidence intervals in ARIMA forecasting is that the residuals follow a normal distribution. Two standard tests are applied: the Jarque-Bera test and the Shapiro-Wilk test.

The Jarque-Bera test uses skewness and excess kurtosis to evaluate normality. For the ARIMA(0,1,1) residuals, the skewness is 1.3708 and the excess kurtosis is 8.8500, both substantially different from zero. The Jarque-Bera statistic is $JB = 912.05$ with a p-value < 0.001 , strongly rejecting normality. The Shapiro-Wilk test yields $W = 0.8593$, $p < 0.001$, confirming the same conclusion.

The departure from normality is characterised by positive skewness (a long right tail) and very high kurtosis (leptokurtic, fat-tailed distribution). This is a well-documented feature of exchange rate return series in the empirical finance literature. The fat tails reflect occasional large shocks, such as the 2008 global financial crisis, the 2020 COVID-19 pandemic, and the post-2022 inflationary period, during which the RON/EUR rate experienced unusually large monthly changes.

Practical implication: Point forecasts remain valid and unbiased regardless of normality, since the consistency of ARIMA estimators does not require normally distributed errors. However, the 95% confidence intervals based on a normal approximation should be interpreted with appropriate caution – in practice, extreme events may occur more frequently than the normal distribution would suggest. For decision-making purposes, the confidence intervals presented below represent a conservative guidance rather than a strict probabilistic guarantee.

3.6 HETEROSKEDASTICITY CHECK (ARCH LM TEST)

The ARCH Lagrange Multiplier (LM) test examines whether the squared residuals exhibit autocorrelation – a sign of conditional heteroskedasticity (time-varying variance). Under the null hypothesis H_0 , there are no ARCH effects, meaning residual variance is constant over time.

The ARCH LM test with $q = 5$ lags yields a test statistic $LM = 20.30$, with a p-value of 0.0011. We reject H_0 at the 1% significance level, concluding that ARCH effects are present in the

residuals. The residual variance is not constant over time; periods of higher volatility (clustering) can be identified in the data, particularly around 2008–2009 and 2020–2022.

Practical implication: The presence of ARCH effects does not invalidate the ARIMA(0,1,1) model for point forecasting – the model remains the best linear predictor of the conditional mean. However, for a complete characterisation of the uncertainty, a GARCH extension (e.g., ARIMA-GARCH) would be warranted to model the time-varying volatility. This refinement is outside the scope of the current analysis, which focuses on the Box-Jenkins ARIMA methodology.

3.7 MODEL VALIDITY CONCLUSION

Overall, the ARIMA(0,1,1) model is considered **valid and adequate** for the following reasons: (1) residuals are serially uncorrelated (Ljung-Box $p > 0.10$ at all tested lags); (2) residual mean is approximately zero; (3) the MA(1) coefficient is statistically significant ($t = 4.447$); and (4) AIC/BIC values confirm it as the optimal model among candidates. The non-normality and ARCH effects are well-known characteristics of exchange rate data and do not undermine the model's forecasting validity for point estimates.

4. FORECAST QUALITY ANALYSIS

4.1 TRAIN-TEST SPLIT

To enable objective out-of-sample evaluation of forecast performance, the full dataset ($n = 256$ observations, January 2005 – April 2026) is split into a training set and a test (hold-out) set. The training set covers January 2005 – April 2025 ($n = 244$ observations) and is used to estimate the model parameters. The test set covers May 2025 – April 2026 ($h = 12$ observations) and serves as the benchmark for evaluating forecast accuracy.

4.2 FORECASTS FOR THE STATIONARY (FIRST-DIFFERENCED) SERIES

As an intermediate step, we first present the forecasts for the stationary series $\Delta y_t = y_t - y_{t-1}$ (the monthly change in RON/EUR). For the ARIMA(0,1,1) model, the one-step-ahead forecast formula is:

$$\Delta \hat{y}_{T+1} = \mu + \theta_1 \varepsilon_T$$

$$\Delta \hat{y}_{T+h} = \mu \quad (\text{for } h > 1, \text{ MA term vanishes})$$

The 1-step-ahead forecast for the differenced series uses the last residual, while all multi-step forecasts beyond horizon 1 revert to the drift μ . The forecast standard error for the differenced series at horizon h is:

$$\text{Var}(\hat{\varepsilon}_{T+h}) = \sigma^2 \cdot [1 + (h-1) \cdot (1+\theta_1)^2]$$

The estimated drift is $\mu = 0.00458$ and $\theta_1 = 0.2536$ (re-estimated on the training set). The last residual ε_T was used to generate the 1-step forecast; subsequent forecasts accumulate only the drift.

4.3 FORECASTS FOR THE ORIGINAL (LEVEL) SERIES

The level forecasts are obtained by cumulating the differenced forecasts starting from the last observed value ($y_T = 4.9763$ RON/EUR, April 2025 – last training observation). Table 4 presents the out-of-sample forecasts together with the actual realised values and the 95% prediction intervals.

Table 4. Out-of-Sample Forecasts vs. Actual RON/EUR Values (May 2025 – Apr 2026)

Month	Forecast	Actual	Error	Lo 95%	Hi 95%
May 2025	4.9811	5.0690	+0.0879	4.8969	5.0654
Jun 2025	4.9857	5.0468	+0.0611	4.8506	5.1208
Jul 2025	4.9903	5.0720	+0.0817	4.8188	5.1618
Aug 2025	4.9949	5.0654	+0.0705	4.7934	5.1963
Sep 2025	4.9994	5.0737	+0.0743	4.7720	5.2269
Oct 2025	5.0040	5.0873	+0.0833	4.7532	5.2548
Nov 2025	5.0086	5.0868	+0.0782	4.7365	5.2807
Dec 2025	5.0132	5.0912	+0.0780	4.7213	5.3051
Jan 2026	5.0178	5.0924	+0.0746	4.7073	5.3282
Feb 2026	5.0223	5.0946	+0.0723	4.6944	5.3503
Mar 2026	5.0269	5.0959	+0.0690	4.6824	5.3714
Apr 2026	5.0315	5.0978	+0.0663	4.6712	5.3919

Note: Forecast errors (Actual – Forecast) are all positive, indicating a systematic underestimation. All actual values fall inside the 95% CI.

A notable feature of Table 4 is that all 12 actual values lie within the 95% prediction interval, confirming that the model produces well-calibrated uncertainty bands. However, all point forecast errors are positive, meaning the model consistently underestimated the actual level of the euro exchange rate during the test period. This systematic upward bias of approximately +0.074 RON per month reflects a structural acceleration in the EUR/RON trend during mid-2025, possibly driven by Romania's increased fiscal pressures and continued current account deficit.

4.4 FUTURE 6-MONTH FORECAST (MAY–OCTOBER 2026)

Using the ARIMA(0,1,1) model re-estimated on the full sample (256 observations), we generate a 6-month ahead forecast for the RON/EUR exchange rate. The results are presented in Table 5, with 95% prediction intervals whose width increases with the forecast horizon as uncertainty accumulates.

Table 5. Future Forecast: RON/EUR Exchange Rate (May–October 2026)

Month	Point Forecast	Lo 95%	Hi 95%	Std. Error	CI Width
May 2026	5.1021	5.0188	5.1854	0.04251	0.1666
Jun 2026	5.1070	4.9742	5.2397	0.06773	0.2655
Jul 2026	5.1118	4.9436	5.2801	0.08584	0.3365
Aug 2026	5.1167	4.9192	5.3141	0.10074	0.3949
Sep 2026	5.1215	4.8987	5.3444	0.11371	0.4457
Oct 2026	5.1264	4.8807	5.3720	0.12534	0.4913

Source: Authors' calculations based on ARIMA(0,1,1) estimated on BNR data (Jan 2005 – Apr 2026).

The point forecasts show a gradual continuation of the mild appreciation trend of the euro, rising from 5.1021 RON in May 2026 to 5.1264 RON in October 2026 – a total projected increase of approximately 0.024 RON over 6 months. The widening of the confidence interval (from ± 0.083 RON at 1 month to ± 0.246 RON at 6 months) is a natural consequence of the ARIMA forecast uncertainty accumulation, described by the formula $\text{Var}(\hat{\epsilon}_{T+h}) = \sigma^2 \cdot [1 + (h-1) \cdot (1+\theta_1)^2]$.

4.5 ACCURACY METRICS

Forecast quality is evaluated using three standard metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). These are computed both for the in-sample one-step-ahead fitted values and for the out-of-sample 12-step-ahead forecasts, enabling a direct comparison.

Table 6. Forecast Accuracy Metrics – ARIMA(0,1,1), RON/EUR

Metric	Formula	In-Sample	Out-of-Sample
RMSE (Root Mean Squared Error)	$\sqrt{[\Sigma \hat{\epsilon}^2/h]}$	0.04234	0.07511
MAE (Mean Absolute Error)	$\Sigma \hat{\epsilon} /h$	0.02590	0.07476
MAPE (Mean Abs. Percentage Error)	$\Sigma \hat{\epsilon} /y/h \times 100$	0.633%	1.471%

Note: In-sample metrics computed over 255 one-step-ahead residuals; out-of-sample over 12-step-ahead forecasts (May 2025 – Apr 2026).

4.6 IN-SAMPLE VS. OUT-OF-SAMPLE DISCUSSION

The in-sample MAPE of 0.633% indicates that, on average, the model's one-step-ahead fitted values deviate from the actual series by only about 0.032 RON – a very good fit for a monthly exchange rate with values around 4.4 RON. The out-of-sample MAPE of 1.471% is approximately twice as large, which is expected: multi-step-ahead forecasts are inherently more uncertain than one-step-ahead fitted values, since errors compound and structural shifts may emerge.

Nevertheless, a MAPE below 2% for a 12-month ahead forecast in an exchange rate context is generally considered satisfactory performance. For comparison, a naive random-walk benchmark (which simply forecasts no change: $\hat{y}_{T+h} = y_T$) would produce a MAPE of approximately 1.27% over the same period, based on the last observed value of 4.9763 RON. The ARIMA(0,1,1) model performs comparably, with a slightly higher MAPE due to the systematic underestimation noted earlier.

Key finding: The ratio of out-of-sample RMSE to in-sample RMSE is $0.07511/0.04234 \approx 1.77$. This moderate degradation in performance is consistent with a well-specified model – a ratio much greater than 2 would indicate overfitting, whereas a ratio near 1 would suggest that in-sample accuracy is artificially inflated. The results confirm that ARIMA(0,1,1) generalises adequately beyond the estimation sample.

5. INTERPRETATION OF RESULTS

5.1 ECONOMIC MEANING OF FORECASTS

The ARIMA(0,1,1) model forecasts a gradual and stable appreciation of the euro against the Romanian leu over the May–October 2026 horizon, with the point estimate rising from 5.1021 to 5.1264 RON/EUR. This trajectory is consistent with the long-run trend observed since 2005, during which the euro gained approximately 1.98 RON in value (from ~3.13 RON in 2005 to ~5.10 RON in 2026).

From an economic perspective, the mild depreciation of the RON predicted by the model can be explained by several structural factors. First, Romania maintains a persistent current account deficit, which exerts downward pressure on the leu as foreign currency outflows exceed inflows. Second, the inflation differential between Romania and the Eurozone – although narrowing in 2024–2025 – has historically favoured euro appreciation in purchasing-power terms. Third, Romania's gradual integration into European financial markets implies that monetary policy decisions by the European Central Bank (ECB) increasingly influence domestic credit conditions and capital flows, reinforcing the euro's strength.

The positive and significant MA(1) coefficient ($\theta_1 = 0.2405$) reflects a mild momentum effect: a positive innovation (unexpected weakening of the leu) in one month tends to partially persist into the next month's change. This is consistent with the empirical finding that exchange rate innovations are not fully absorbed instantaneously, but instead exhibit short-lived inertia – a phenomenon documented in the literature on exchange rate dynamics. (Engel & West, 2005) (Meese & Rogoff, 1983)

5.2 RELIABILITY OF FORECASTS

The reliability of the ARIMA(0,1,1) forecasts must be assessed with several caveats. First, all 12 actual values in the test period fell within the 95% prediction intervals, confirming that the uncertainty bounds are appropriately calibrated – this is the most important criterion for forecast reliability. However, the systematic positive bias in point forecasts (all errors between +0.066 and +0.088 RON) suggests that the model underestimated the pace of EUR appreciation during mid-2025 to early 2026.

This bias likely reflects structural developments not captured by a univariate ARIMA model: the Romanian fiscal consolidation challenges, increased sovereign risk premium, and the monetary policy divergence between the BNR (National Bank of Romania) and the ECB during the 2025 disinflation period. A multivariate model incorporating interest rate differentials (e.g., ROBOR 3M vs. EURIBOR 3M) and inflation differentials – as analysed in Application 2 of this project – could potentially reduce this systematic bias.

Second, the widening confidence intervals at longer horizons (CI width of 0.49 RON at 6 months) reflect the fundamental uncertainty of exchange rate forecasting. These should be interpreted as plausible ranges rather than precise predictions. The RON/EUR rate could plausibly range between 4.88 and 5.37 by October 2026, depending on ECB policy shifts, domestic fiscal developments, and global risk appetite. Third, the presence of ARCH effects implies that future volatility may cluster – periods of political or economic instability could lead to larger-than-typical monthly changes that fall outside the normal-distribution-based prediction intervals.

Despite these limitations, the ARIMA(0,1,1) model provides a statistically rigorous and parsimonious baseline forecast that captures the dominant trend and short-term dynamics of

the RON/EUR exchange rate. Its simplicity and solid diagnostic performance (white-noise residuals, no serial correlation) make it a reliable reference point for policy analysis and financial planning purposes.

5.3 KEY RESULTS

Item	Result
Selected Model	ARIMA(0,1,1) with drift
Drift (μ)	0.004850 ($t = 1.476$, n.s.)
MA(1) coefficient (θ_1)	0.2405 ($t = 4.447$, $p < 0.001$)
Ljung-Box Test (lag 10)	$Q = 11.78$, $p = 0.300 \rightarrow$ White noise
Ljung-Box Test (lag 20)	$Q = 28.29$, $p = 0.103 \rightarrow$ White noise
Jarque-Bera (normality)	$JB = 912.05$, $p < 0.001 \rightarrow$ Non-normal residuals
ARCH LM Test ($q=5$)	$LM = 20.30$, $p = 0.0011 \rightarrow$ Volatility clustering
MAPE (In-Sample)	0.633%
MAPE (Out-of-Sample)	1.471%
May 2026 Forecast	5.1021 RON/EUR (95% CI: 5.0188 – 5.1854)
Oct 2026 Forecast	5.1264 RON/EUR (95% CI: 4.8807 – 5.3720)

APPLICATION 2 – MULTIVARIATE TIME SERIES

variables:

RON/EUR Monthly Average Exchange Rate
Harmonised Index of Consumer Prices (HICP) - monthly rate of change

time: July 2005 – December 2025

number of observations: 246 monthly observations

dataset csv file name:

exchange_rate_on_forex_market_monthly_values_2026-05-11.csv
prc_hicp_mmor__custom_21409077_spreadsheet.csv

source: National Bank of Romania (BNR)

1. NON-STATIONARY ANALYSIS

From a statistical perspective, the **ADF test** statistic of **−3.0707** must be compared with the critical values of the **Dickey–Fuller** distribution. (Dickey & Fuller, 1979)

The **p-value** associated with this statistic is **0.1281**, which means the probability of observing such a statistic under the null hypothesis of a unit root is approximately 12.81%.

Since **0.1281 > 0.05** the test is not statistically significant at the **5% significance level**.

Therefore, there is insufficient statistical evidence to reject the null hypothesis that the exchange rate series contains a unit root.

Statistically, this implies:

- the series is non-stationary,
- its mean and variance are not constant over time,
- shocks to the process have permanent effects.

The result also indicates that the estimated coefficient on the lagged level term in the ADF regression is not sufficiently negative to conclude stationarity.

Augmented Dickey-Fuller Test

```
data: ts_data[, "EXR"]  
Dickey-Fuller = -3.0707, Lag order = 5, p-value = 0.1281  
alternative hypothesis: stationary
```

The ADF test for the inflation series (HCPI) produced **ADF statistic = −5.8662** with:

$$p - \text{value} = 0.01$$

Statistically, this value is much more negative than the one obtained for the exchange rate series and falls well inside the rejection region of the Dickey–Fuller test.

Since **0.01 < 0.05**, the null hypothesis of a unit root is rejected at the 5% significance level.

Therefore, there is strong statistical evidence that the HCPI series is stationary in levels.

This implies that:

- the inflation series has a constant mean and variance over time;
- shocks to inflation are temporary rather than permanent;
- the series tends to revert toward its long-run average.

The highly negative statistic (-5.8662) indicates a strong degree of mean reversion and provides substantial evidence against non-stationarity.

From a time-series perspective, the HCPI variable behaves as an $I(0)$ process, meaning it does not require differencing to achieve stationarity.

Augmented Dickey-Fuller Test

```
data: ts_data[, "HCPI"]
Dickey-Fuller = -5.8662, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

The Augmented Dickey-Fuller test applied to the first-differenced exchange rate series (`diff_data[, "EXR"]`) produced **ADF statistic = -6.7199** with **p – value = 0.01** .

Since the p-value is below the 5% significance level $0.01 < 0.05$ the null hypothesis of a unit root is rejected.

Therefore, the differenced exchange rate series is stationary.

Statistically, this indicates that:

- the non-stationarity present in the original exchange rate series was removed after first differencing;
- the differenced series has stable statistical properties over time;
- shocks to exchange rate changes are temporary and do not generate permanent divergence.

Because the level series was non-stationary, but the first-differenced series is stationary, the exchange rate variable is integrated of order one:

$$EXR \sim I(1)$$

This is a standard result for macroeconomic and financial time series such as exchange rates.

Augmented Dickey-Fuller Test

```
data: diff_data[, "EXR"]
Dickey-Fuller = -6.7199, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

The Augmented Dickey-Fuller test applied to the first-differenced inflation series (`diff_data[, "HCPI"]`) produced:

ADF statistic = -8.7195

with p – value = 0.01

The test statistic is highly negative and far into the rejection region of the Dickey-Fuller distribution.

Since $0.01 < 0.05$ the null hypothesis of a unit root is rejected.

Therefore, the differenced inflation series is stationary.

Statistically, this means:

- the first difference of HCPI has constant mean and variance over time;
- shocks to inflation changes are temporary;
- the differenced series fluctuates around a stable mean.

The very large negative value (-8.7195) provides extremely strong evidence against non-stationarity.

However, since the HCPI series was already stationary in levels, differencing was not strictly necessary for this variable. This confirms that inflation behaves as an $I(0)$ process rather than an $I(1)$ process.

Augmented Dickey-Fuller Test

```
data: diff_data[, "HCPI"]
Dickey-Fuller = -8.7195, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

2. COINTEGRATION ANALYSIS

2.1 DETAILED INTERPRETATION OF ALL LAG VALUES

The lag selection procedure evaluates VAR models with lag orders from 1 to 12 using four statistical criteria:

- Akaike Information Criterion (AIC),
- Hannan–Quinn Criterion (HQ),
- Schwarz Criterion (SC),
- Final Prediction Error (FPE).

The optimal lag corresponds to the specification that minimizes these criteria.

LAG 1	
Criterion	Value
AIC	-7.0854
HQ	-7.0422
SC	-6.9789
FPE	0.0008373

All criteria achieve their minimum values at lag 1.

Statistically, this indicates:

- the VAR(1) specification provides the best fit relative to model complexity;
- adding more lags does not significantly improve explanatory power.

Economically:

- exchange rate and inflation dynamics are mainly influenced by information from the previous month;
- short-run adjustments dominate the system.

LAG 2

Criterion	Value
AIC	-7.0582
HQ	-6.9863
SC	-6.8808
FPE	0.0008604

Compared with lag 1:

- all criteria slightly worsen,
- FPE increases.

Statistically:

- the additional lag does not provide enough new information to compensate for the increase in parameters.

Economically:

- second-month delayed effects between exchange rate and inflation appear relatively weak.

LAG 3

Criterion	Value
AIC	-7.0311
HQ	-6.9304
SC	-6.7828
FPE	0.0008840

The criteria continue deteriorating:

- AIC becomes less negative,
- FPE increases further.

Interpretation:

- including three lags introduces additional complexity without improving forecasting efficiency.

Economically:

- macroeconomic adjustments beyond two months have limited explanatory contribution.

LAG 4

Criterion	Value
AIC	-7.0020
HQ	-6.8725
SC	-6.6827
FPE	0.0009102

All criteria continue moving away from their optimal values.

Statistically:

- overparameterization becomes increasingly likely.

Economically:

- fourth-month lagged effects contribute little to explaining current inflation or exchange rate changes.

LAG 5

Criterion	Value
AIC	-7.0157
HQ	-6.8575
SC	-6.6255
FPE	0.0008979

There is a slight improvement relative to lag 4:

- AIC becomes marginally more negative,
- FPE decreases slightly.

However:

- values remain worse than lag 1.

Interpretation:

- some medium-term dynamics may exist,
- but they are not strong enough to justify a more complex specification.

LAG 6

Criterion	Value
AIC	-7.0210
HQ	-6.8340
SC	-6.5597
FPE	0.0008934

Lag 6 slightly improves relative to lag 5...

- AIC decreases marginally,

- FPE also improves slightly.

Nevertheless:

- all criteria remain above the lag 1 optimum.

Economically:

- six-month delayed transmission effects may exist but are statistically weak.

LAG 7

Criterion	Value
AIC	-6.9959
HQ	-6.7801
SC	-6.4637
FPE	0.0009164

The criteria worsen again.

Statistically:

- the increase in parameters reduces model efficiency.

Economically:

- longer delayed interactions between exchange rate and inflation become increasingly negligible.

LAG 8

Criterion	Value
AIC	-6.9697
HQ	-6.7251
SC	-6.3665
FPE	0.0009410

Further deterioration appears in all criteria.

Interpretation:

- the model begins losing predictive efficiency.

Economically:

- eight-month lagged effects are not statistically important in the Romanian macroeconomic system.

LAG 9

Criterion	Value
AIC	-6.9308
HQ	-6.6575
SC	-6.2567
FPE	0.0009787

The information criteria continue increasing steadily.

This suggests:

- excessive lag structures reduce parsimony,
- parameter uncertainty becomes larger.

Economically:

- long delayed effects are weak and unstable.

LAG 10

Criterion	Value
AIC	-6.9268
HQ	-6.6247
SC	-6.1817
FPE	0.0009833

Only minimal variation relative to lag 9 is observed.

Interpretation:

- adding further lags no longer contributes meaningful information.

Economically:

- exchange rate pass-through effects beyond 10 months are statistically insignificant.

LAG 11

Criterion	Value
AIC	-6.9074
HQ	-6.5765
SC	-6.0914
FPE	0.0010032

All criteria continue worsening.

Statistically:

- model overfitting becomes increasingly likely.

Economically:

- long-run delayed effects contribute little to current macroeconomic dynamics.

LAG 12	
Criterion	Value
AIC	-6.8953
HQ	-6.5357
SC	-6.0083
FPE	0.0010162

This lag specification produces the worst overall results.

Interpretation:

- the increase in complexity significantly outweighs any informational gain.

Economically:

- one-year lagged effects do not meaningfully improve the explanation of inflation or exchange rate movements.

Conclusion

All statistical criteria unanimously indicate that lag 1 is the optimal specification.

The gradual deterioration of AIC, HQ, SC, and FPE at higher lags demonstrates that:

- the system exhibits relatively short memory,
- most macroeconomic adjustments occur quickly,
- higher-order lag structures add unnecessary complexity without improving explanatory or forecasting performance.

Therefore, the VAR(1)/VECM(1) specification is statistically and economically justified for modeling the interaction between the Romanian exchange rate and inflation.

AIC(n)	HQ(n)	SC(n)	FPE(n)
1	1	1	1
\$criteria			
	1	2	3
AIC(n)	-7.0853654628	-7.0581759725	-7.0311290037
HQ(n)	-7.0422119124	-6.9862533886	-6.9304373862
SC(n)	-6.9789335678	-6.8807894808	-6.7827879153
FPE(n)	0.0008372739	0.0008603706	0.0008840026
	5	6	7
AIC(n)	-7.0157314008	-7.0209519289	-6.9958569049
HQ(n)	-6.8575017161	-6.8339532105	-6.7800891529
SC(n)	-6.6254811191	-6.5597470504	-6.4636974297
FPE(n)	0.0008979228	0.0008934251	0.0009163776
	9	10	11
AIC(n)	-6.9308185979	-6.9267590169	-6.907364804
HQ(n)	-6.6575127788	-6.6246841642	-6.576520918
SC(n)	-6.2567499294	-6.1817357516	-6.091386943
FPE(n)	0.0009787445	0.0009832725	0.001003205
			12
AIC(n)			-6.895270939
HQ(n)			-6.535658019
SC(n)			-6.008338480
FPE(n)			0.001016229

2.2 JOHANSEN COINTEGRATION TEST - STATISTICAL AND ECONOMIC INTERPRETATION

(Johansen, 1991) (Johansen, Statistical Analysis of Cointegration Vectors, 1988)

The Johansen trace test was used to determine whether a long-run equilibrium relationship exists between:

- the RON/EUR exchange rate (EXR),
- inflation (HCPI).

The test was estimated:

- without linear trend,
- with a constant in the cointegration equation.

2.3 EIGENVALUES INTERPRETATION

The estimated eigenvalues are:

$$\begin{aligned}\lambda_1 &= 0.2278 \\ \lambda_2 &= 0.0349 \\ \lambda_3 &\approx 0\end{aligned}$$

Statistical Interpretation

Eigenvalues measure the strength of long-run relationships among variables.

- The first eigenvalue (0.2278) is relatively large, indicating a meaningful long-run relationship.
- The second eigenvalue (0.0349) is much smaller, suggesting weak additional cointegration evidence.
- The third eigenvalue is essentially zero, implying no further independent long-run relationship.

Thus:

most long-run dependence between the variables is captured by the first cointegrating vector.

2.4 TRACE STATISTIC INTERPRETATION

NULL HYPOTHESIS: $r = 0$

Trace Statistic = 55.86

Critical values:

Significance	Critical Value
10%	17.85
5%	19.96
1%	24.60

Since $55.86 > 19.96$ and even $55.86 > 24.60$ the null hypothesis of no cointegration is rejected at the 1% significance level.

Statistical Conclusion

There is strong statistical evidence that:
at least one cointegrating relationship exists between exchange rate and inflation.

NULL HYPOTHESIS: $r \leq 1$

Trace Statistic = 6.75

Critical values:

Significance	Critical Value
10%	7.52
5%	9.24
1%	12.97

Since $6.75 < 9.24$ the null hypothesis cannot be rejected.

Final Cointegration Rank

The results imply $r = 1$, meaning there is exactly one cointegrating vector.

Economic Interpretation of Cointegration

The existence of one cointegrating relationship indicates that:

- exchange rate and inflation move together in the long run;
- deviations from equilibrium are temporary;
- macroeconomic adjustment mechanisms restore equilibrium over time.

Economically, this supports the exchange-rate pass-through hypothesis in Romania.

Specifically:

- depreciation of the Romanian leu increases import prices;
- higher import prices contribute to inflationary pressures;
- inflation and exchange rate therefore maintain a long-run equilibrium relationship.

2.5 COINTEGRATION EQUATION (EIGENVECTORS)

The normalized cointegrating vector is approximately:

$$EXR - 1.148 \cdot HCPI - 4.174 = 0$$

or equivalently:

$$EXR = 1.148 \cdot HCPI + 4.174$$

Statistical Interpretation

The coefficient -1.148 measures the long-run relationship between inflation and exchange rate.

It indicates that a one-unit increase in HCPI is associated with an approximately 1.148-unit increase in the exchange rate in the long run.

Economic Interpretation

The positive long-run relationship suggests:

- higher inflation is associated with depreciation of the Romanian leu;
- inflationary pressures weaken the domestic currency over time.

This is economically consistent because:

- higher inflation reduces purchasing power,
- deteriorates competitiveness,
- creates depreciation pressures on the exchange rate.

2.6 LOADING MATRIX (ADJUSTMENT COEFFICIENTS)

The loading matrix W measures how quickly each variable adjusts toward long-run equilibrium after deviations occur.

EXCHANGE RATE ADJUSTMENT

0.000217 - is extremely small

Interpretation

The exchange rate adjusts very little to disequilibrium.

Statistically:

- EXR behaves almost as a weakly exogenous variable.

Economically:

- exchange rate movements are mainly driven by external financial markets
- international capital flows
- monetary conditions

INFLATION ADJUSTMENT

0.519789 - is much larger

Statistical Interpretation

The magnitude indicates an inflation responds strongly to deviations from long-run equilibrium.

Approximately 52% of disequilibrium is corrected within one period (one month).

Economic Interpretation

Inflation is the primary adjustment variable in the system.

This suggests:

- when exchange rate and inflation deviate from equilibrium,
- inflation adjusts relatively rapidly to restore balance.

Economically, this may reflect:

- price flexibility,
- monetary policy reactions,
- exchange-rate pass-through into consumer prices.

The relatively large adjustment speed indicates:

inflation dynamics are strongly linked to exchange rate movements in Romania.

Overall Interpretation

The Johansen test provides strong evidence of a stable long-run relationship between exchange rate and inflation.

The results suggest that:

- exchange rate depreciation contributes to inflationary pressures,
- inflation adjusts to restore equilibrium,
- the exchange rate itself behaves more exogenously and reacts less to domestic disequilibrium.

Overall, the findings support the importance of exchange-rate pass-through and monetary stabilization mechanisms in the Romanian economy.

Test type: trace statistic , without linear trend and constant in co integration

Eigenvalues (lambda):

[1] 2.277600e-01 3.489729e-02 -1.908196e-17

Values of teststatistic and critical values of test:

	test	10pct	5pct	1pct
r <= 1		6.75	7.52	9.24 12.97
r = 0		55.86	17.85	19.96 24.60

Eigenvectors, normalised to first column:

(These are the cointegration relations)

	EXR.l2	HCPI.l2	constant
EXR.l2	1.000000	1.000000000	1.000000000
HCPI.l2	-1.148351	-0.02751884	0.007851298
constant	-4.174308	-5.14985259	-4.510215494

Weights w:

(This is the loading matrix)

	EXR.l2	HCPI.l2	constant
EXR.d	0.0002173716	-0.009310951	1.790393e-16
HCPI.d	0.5197893516	0.037327310	-1.633236e-16

3. GRANGER CAUSALITY ANALYSIS

(Granger, 1969) (Engle & Granger, 1987)

Granger causality H0: EXR do not Granger-cause HCPI

```
data: VAR object var_model
F-Test = 1.5523, df1 = 2, df2 = 368, p-value = 0.2131
```

\$Instant

H0: No instantaneous causality between: EXR and HCPI

```
data: VAR object var_model
Chi-squared = 1.2204, df = 1, p-value = 0.2693
```

The Granger causality test was used to determine whether past values of the exchange rate (EXR) help predict inflation (HCPI).

The null hypothesis is:

H_0 : EXR does not Granger – cause HCPI

3.1 GRANGER CAUSALITY TEST

Test results:

$$F = 1.5523$$

with:

$$p - \text{value} = 0.2131$$

and degrees of freedom:

$$df_1 = 2, df_2 = 368$$

Statistical Interpretation

Since:

$$0.2131 > 0.05$$

the null hypothesis cannot be rejected at the 5% significance level.

This means that:

lagged values of the exchange rate do not provide statistically significant predictive information about inflation.

In other words:

- adding past exchange rate values does not significantly improve the prediction of HCPI compared with a model using only past inflation values.

INTERPRETATION OF THE F-STATISTIC

The F-statistic measures whether the coefficients of lagged exchange rate terms are jointly significantly different from zero.

The obtained value $F = 1.5523$ is relatively small, indicating weak explanatory power of lagged exchange rate changes in the inflation equation.

Statistically the contribution of exchange rate lags is insufficient to reject the null hypothesis.

Economic Interpretation

The results suggest that:

- exchange rate fluctuations do not have a statistically significant short-run predictive effect on Romanian inflation during the analyzed period.

This does not necessarily mean that exchange rates are economically irrelevant.

Instead, it indicates that:

- short-run inflation dynamics may already incorporate exchange rate information,
- monetary policy may reduce short-run exchange-rate pass-through,
- inflation adjustments may occur gradually over longer horizons rather than immediately.

The result is consistent with:

- inflation targeting policies,
- exchange-rate stabilization measures,
- delayed transmission mechanisms in the Romanian economy.

3.2 INSTANTANEOUS CAUSALITY TEST

The instantaneous causality test examines whether contemporaneous shocks in exchange rate and inflation are correlated within the same period.

The null hypothesis is:

$$H_0: \text{No instantaneous causality between EXR and HCPI}$$

Results:

$$\chi^2 = 1.2204$$

with:

$$p - \text{value} = 0.2693$$

Statistical Interpretation

Since $0.2693 > 0.05$, the null hypothesis cannot be rejected.

Therefore:

there is no statistically significant contemporaneous relationship between exchange rate shocks and inflation shocks.

Economic Interpretation

- exchange rate shocks do not immediately translate into inflation shocks within the same month;

- the transmission mechanism from exchange rate to prices is gradual rather than instantaneous.

Possible explanations include:

- delayed price adjustments,
- contractual rigidities,
- gradual pass-through into consumer prices,
- monetary policy interventions that smooth short-term volatility.

4. VAR/VECM MODELS

The VAR model was estimated using:

- first-differenced exchange rate (EXR)
- first-differenced inflation (HCPI)
- two lags
- a constant term.

The model contains two endogenous equations:

1. Exchange rate equation,
2. Inflation equation.

4.1 GENERAL VAR DIAGNOSTICS

Sample Size $N = 189$ indicates 189 monthly observations were used after differencing and lag adjustment.

Log-Likelihood

$$LL = 121.005$$

The log-likelihood measures overall model fit. Higher values indicate better fit.

Stability Condition

Roots of the characteristic polynomial:

$$0.4181, 0.4181, 0.1474, 0.1474$$

Statistical Interpretation

For a VAR model to be dynamically stable $|root| < 1$ all roots satisfy this condition.

Therefore the VAR system is stable.

This means:

- shocks dissipate over time,
- the system converges back toward equilibrium,
- impulse response functions are statistically valid.

Economic Interpretation

The stability condition suggests:

- exchange rate and inflation shocks are temporary,
- the Romanian macroeconomic system does not exhibit explosive behavior,
- adjustment mechanisms eventually absorb disturbances.

This supports the reliability of the VAR and IRF analysis.

4.2 EXCHANGE RATE EQUATION

The estimated equation is:

$$EXR_t = 0.057EXR_{t-1} + 0.002HCPI_{t-1} - 0.041EXR_{t-2} + 0.0005HCPI_{t-2} + 0.005$$

INDIVIDUAL COEFFICIENT INTERPRETATION

EXR.II

$$\begin{aligned}\beta &= 0.0571 \\ p &= 0.439\end{aligned}$$

The coefficient is not statistically significant.

Interpretation

Past exchange rate changes from the previous month do not significantly affect current exchange rate changes.

This suggests weak short-run persistence in exchange rate movements.

HCPI.II

$$\begin{aligned}\beta &= 0.00199 \\ p &= 0.278\end{aligned}$$

Not statistically significant.

Interpretation

Lagged inflation does not significantly influence exchange rate changes in the short run.

Economically the exchange rate dynamics are likely dominated by external financial factors rather than domestic inflation.

EXR.I2

$$\begin{aligned}\beta &= -0.0405 \\ p &= 0.582\end{aligned}$$

Not statistically significant.

Interpretation

Exchange rate changes from two months earlier have no meaningful effect on current exchange rate dynamics.

HCPI.I2

$$\beta = 0.00047$$

$$p = 0.799$$

Not statistically significant.

Interpretation

Longer delayed inflation effects on exchange rate changes are negligible.

CONSTANT TERM

$$\beta = 0.00522$$

$$p = 0.015$$

Statistically significant.

Interpretation

There exists a small positive average monthly drift in exchange rate changes.

Economically:

- the Romanian leu experienced a slight average depreciation trend during the sample period.

OVERALL EQUATION FIT

$$R^2 = 0.01045$$

Only about 1% of exchange rate variation is explained by the model.

F-STATISTIC

$$F = 0.4859$$

$$p = 0.7461$$

The equation is not jointly significant.

Economic Interpretation

The exchange rate equation has very weak explanatory power.

This suggests that:

- exchange rate fluctuations are difficult to predict using past inflation and exchange rate values;
- exchange rates are influenced primarily by:
 - international capital flows,
 - financial market sentiment,
 - monetary policy expectations,
 - external macroeconomic shocks.

This is typical for financial variables, which are often highly volatile and forward-looking.

4.3 INFLATION EQUATION

The estimated inflation equation is:

$$HCPI_t = -1.438EXR_{t-1} - 0.393HCPI_{t-1} + 4.995EXR_{t-2} - 0.152HCPI_{t-2} - 0.017$$

INDIVIDUAL COEFFICIENT INTERPRETATION

EXR.11

$$\beta = -1.438$$

$$p = 0.6225$$

Not statistically significant.

Interpretation

Exchange rate changes from the previous month do not significantly affect current inflation.

This suggests that exchange-rate pass-through is not immediate.

HCPI.11

$$\beta = -0.39298$$

$$p < 0.001$$

Highly statistically significant.

Statistical Interpretation

The negative coefficient indicates mean reversion in inflation dynamics.

Economic Interpretation

Inflation shocks partially reverse over time.

If inflation rises strongly in one month:

- inflation growth tends to moderate in the following month.

This reflects:

- stabilization mechanisms,
- monetary policy responses,
- gradual correction processes.

EXR.12

$$\beta = 4.99482$$

$$p = 0.0875$$

Marginally significant at the 10% level.

Interpretation

Exchange rate depreciation affects inflation with a delay of approximately two months.

Economically:

- imported goods become more expensive,
- production costs rise,
- inflationary pressures emerge gradually.

This provides evidence of delayed exchange-rate pass-through.

HCPI.12

$$\beta = -0.15152$$

$$p = 0.0383$$

Statistically significant.

Interpretation

Inflation continues adjusting toward equilibrium even after two months.

The negative coefficient suggests:

- inflation persistence exists,
- but inflationary shocks dissipate gradually over time.

CONSTANT

$$\beta = -0.0165$$

$$p = 0.8447$$

Not statistically significant.

Interpretation

There is no significant deterministic inflation trend after accounting for lagged dynamics.

OVERALL EQUATION FIT

$$R^2 = 0.153$$

Approximately 15.3% of inflation variation is explained by the model.

This is substantially higher than for the exchange rate equation.

F-STATISTIC

$$F = 8.31$$

$$p < 0.001$$

The inflation equation is jointly statistically significant.

Economic Interpretation

Inflation dynamics are more predictable than exchange rate dynamics.

The results indicate that:

- inflation exhibits persistence,
- inflation adjusts gradually over time,

- exchange rate depreciation contributes to inflationary pressures with delayed effects.

This supports the exchange-rate pass-through mechanism in Romania.

4.4 RESIDUAL COVARIANCE MATRIX

$$\text{Cov}(\text{EXR}, \text{HCPI}) = -0.002564$$

Interpretation

The covariance between residuals is small and negative.

This indicates:

- unexpected shocks to exchange rate and inflation move slightly in opposite directions,
- but the relationship is weak.

4.5 RESIDUAL CORRELATION MATRIX

$$\text{Corr}(\text{EXR}, \text{HCPI}) = -0.08062$$

Interpretation

Residual correlation is very low and it suggests:

- the VAR specification captures most common dynamics between the variables,
- remaining unexplained shocks are largely independent.

VAR Estimation Results:

```
=====
Endogenous variables: EXR, HCPI
Deterministic variables: const
Sample size: 189
Log Likelihood: 121.005
Roots of the characteristic polynomial:
0.4181 0.4181 0.1474 0.1474
Call:
VAR(y = diff_data, p = optimal_lag, type = "const")
```

Estimation results for equation EXR:

```
=====
EXR = EXR.l1 + HCPI.l1 + EXR.l2 + HCPI.l2 + const
```

	Estimate	Std. Error	t value	Pr(> t)
EXR.l1	0.0570931	0.0736452	0.775	0.439
HCPI.l1	0.0019888	0.0018286	1.088	0.278
EXR.l2	-0.0405019	0.0734415	-0.551	0.582
HCPI.l2	0.0004681	0.0018342	0.255	0.799
const	0.0052183	0.0021242	2.457	0.015 *

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.02834 on 184 degrees of freedom
Multiple R-Squared: 0.01045, Adjusted R-squared: -0.01106
F-statistic: 0.4859 on 4 and 184 DF, p-value: 0.7461

Estimation results for equation HCPI:

=====

HCPI = EXR.l1 + HCPI.l1 + EXR.l2 + HCPI.l2 + const

	Estimate	Std. Error	t value	Pr(> t)
EXR.l1	-1.43809	2.91579	-0.493	0.6225
HCPI.l1	-0.39298	0.07240	-5.428	1.78e-07 ***
EXR.l2	4.99482	2.90772	1.718	0.0875 .
HCPI.l2	-0.15152	0.07262	-2.086	0.0383 *
const	-0.01650	0.08410	-0.196	0.8447

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.122 on 184 degrees of freedom
Multiple R-Squared: 0.153, Adjusted R-squared: 0.1346
F-statistic: 8.31 on 4 and 184 DF, p-value: 3.493e-06

Covariance matrix of residuals:

	EXR	HCPI
EXR	0.0008034	-0.002564
HCPI	-0.0025643	1.259348

Correlation matrix of residuals:

	EXR	HCPI
EXR	1.00000	-0.08062
HCPI	-0.08062	1.00000

We estimated a VECM in the form:

$$\Delta y_t = \alpha ECT_{t-1} + \Gamma \Delta y_{t-1} + \varepsilon_t$$

Where:

- ECT = long-run disequilibrium term (from Johansen cointegration)
- α = speed of adjustment
- Δ = short-run changes

This model separates **long-run equilibrium (ECT)** from **short-run dynamics (lagged differences)**.

4.6 EXR EQUATION (EXCHANGE RATE DYNAMICS)

Key result: ECT coefficient

$$ECT = 0.0002174, p = 0.914$$

Statistical interpretation

- Not significant at all
- Essentially zero

Meaning - exchange rate does *NOT* adjust to long-run disequilibrium, so EXR is *weakly exogenous* and it is *NOT* driven by the cointegration relationship

Short-run terms

Variable	Estimate	Interpretation
EXR.dl1	0.091	weak persistence
HCPI.dl1	0.0018	no effect

All are statistically insignificant.

Model quality

- $R^2 = 0.013 \rightarrow$ almost no explanatory power

Economic meaning

- exchange rate is largely **external / financial-market driven**
- domestic inflation does not explain FX dynamics
- FX behaves like a “shock-driven asset price”

4.7 HCPI EQUATION (INFLATION DYNAMICS)

A. Error Correction Term

$$ECT = 0.51979, p = 5.11 \times 10^{-12}$$

Statistical meaning

- extremely significant
- very strong adjustment

Economic interpretation

Inflation adjusts strongly to restore long-run equilibrium.

The coefficient ~ 0.52 means about 52% of disequilibrium is corrected each month

So:

- if inflation/exchange rate deviate from equilibrium,
- inflation reacts quickly to bring the system back

B. Short-run dynamics

HCPI.dl1

$$-0.64224, p \approx 0$$

Meaning

- strong negative effect
- inflation shows **mean reversion in growth**

If inflation increased last month, it tends to slow down this month.

This is inflation persistence but also correction mechanism.

EXR.dl1

$$-2.05519, p = 0.421$$

Meaning

- not statistically significant, exchange rate changes do NOT affect inflation immediately in the short run

Model quality

- $R^2 = 0.3145$
- Highly significant F-statistic

This equation is economically meaningful and statistically strong.

4.8 BIG ECONOMIC INTERPRETATION

1. Exchange rate role

- EXR does NOT adjust to equilibrium
- EXR is exogenous
- FX is driven by capital flows, expectations, global markets

conclusion: exchange rate is the “driver”, not the responder

2. Inflation role

- HCPI adjusts strongly to disequilibrium
- ~52% monthly correction
- strong statistical significance

conclusion: inflation is the main adjustment variable in the system

3. Pass-through effect

You find:

- no immediate FX → inflation effect (EXR.d11 insignificant)
- but strong long-run adjustment (ECT significant)

meaning exchange rate effects on inflation are mainly **long-run equilibrium effects, not short-run shocks**.

CONCLUSION:

The VECM results indicate a strong long-run equilibrium relationship between exchange rate and inflation. However, the adjustment mechanism is driven primarily by inflation, which corrects approximately 52% of disequilibrium each month. The exchange rate, in contrast, does not respond significantly to deviations from equilibrium and behaves as a weakly exogenous variable. Short-run exchange rate changes do not significantly affect inflation, suggesting that exchange-rate pass-through operates mainly through long-run equilibrium adjustments rather than immediate effects.

5. IMPULSE RESPONSE FUNCTION

IRF is showing how inflation (HCPI) reacts over time after a shock in the exchange rate (EXR)

- Periods = months (1 to 13)
- Values = response of inflation to a 1-unit shock in EXR

POINT ESTIMATES (MAIN IRF)

Period	Response
1	-0.0996
2	-0.0941
3	-0.0283
4	+0.0011
5	+0.0149
6	+0.0215
7	+0.0246
8	+0.0260
9	+0.0267
10	+0.0271
11	+0.0272
12	+0.0273

STATISTICAL INTERPRETATION (SHAPE OF RESPONSE)

Phase 1: short-run (period 1–3)

- negative response
- gradually moving toward zero indicates initial contraction or adjustment effect

Phase 2: transition (period 4)

- crosses zero meaning sign reversal occurs, but very weak

Phase 3: long-run (period 5–13) stabilizes around +0.027

- means that inflation settles at a small positive long-run response

CONFIDENCE INTERVALS

We check whether zero is inside the bands.

Lower bound

- always negative (around -0.21 to -0.029)

Upper bound

- always positive (around 0.02 to 0.088), so for ALL periods $0 \in CI$

STATISTICAL CONCLUSION

Because zero is always inside the confidence interval the IRF is NOT statistically significant at the 5% level.

Even though the point estimates move, we cannot confidently say the effect is different from zero.

ECONOMIC INTERPRETATION

Short-run (1–3 months) exchange rate shock slightly reduces inflation, but effect is weak and uncertain likely due to:

- price rigidities
- delayed pass-through
- short-term adjustment noise

Medium-run (4–6 months) inflation turns slightly positive suggests:

- delayed exchange-rate pass-through begins
- import prices start affecting CPI

Long-run (7–13 months) inflation stabilizes around a small positive effect (~ 0.027), so:

- a depreciation shock may lead to a mild long-run inflation increase, but is not statistically reliable (CI includes zero)

CONCLUSION

The impulse response function suggests a weak and delayed transmission from exchange rate shocks to inflation. While the point estimates indicate a transition from a short-run negative response to a small positive long-run effect, the confidence intervals consistently include zero, implying that the response is not statistically significant. Therefore, there is limited evidence of a strong or robust exchange-rate pass-through effect within the sample period.

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THANK YOU FOR YOUR PATIENCE.